Ummed Singh
MTS@ByteXL | Ex - Nagarro | Java |
Spring Boot | Microservices | React...

#15 Swagger/OpenAPI Documentation

Swagger/OpenAPI Documentation	API documentation, endpoint testing, Swagger UI configuration

API Documentation

API Documentation is a document that explains:

- What APIs are available
- What request data should be sent
- What response data will be returned
- Authentication requirements
- Error responses
- API examples

Without documentation:

- **Frontend Developer:** "Which API should I call?"
- **Backend Developer:** "Check the code."
- **With Swagger/OpenAPI:**

  Frontend Developer:
  Open Swagger UI
  → See all APIs
  → Test APIs directly
  → View request & response format

OpenAPI

OpenAPI is a specification (standard format) for describing REST APIs.

It defines:

- Endpoints
- Methods
- Request Parameters

- Request Body
- Responses
- Authentication

Example OpenAPI Description:

POST /students

Request:

```
{
 "name":"Ummed Singh",
 "email":"ummed@gmail.com"
}
```

Response:

```
{
 "id":1,
 "name":"Ummed Singh",
 "email":"ummed@gmail.com"
}
```

Swagger

Swagger is a tool that implements the OpenAPI specification.

It provides:

Swagger UI: Interactive webpage where APIs can be tested.

Swagger Documentation: Automatically generated API documentation.

API Testing: No need for Postman for quick testing.

Ummed Singh
MTS@ByteXL | Ex - Nagarro | Java | Spring Boot | Microservices | React...

OpenAPI vs Swagger

OpenAPI	Swagger
Specification	Tool
Standard Format	Implementation
Defines API Structure	Generates UI
API Contract	API Visualization

Example

Suppose you're creating a Student Management System.

APIs:

POST /students
GET /students
GET /students/{id}
PUT /students/{id}
DELETE /students/{id}

Swagger automatically generates:

Student Controller

POST Create Student
GET Get All Students
GET Get Student By ID
PUT Update Student
DELETE Delete Student

Spring Boot Swagger Setup

Dependency

Ummed Singh
MTS@ByteXL | Ex - Nagarro | Java |
Spring Boot | Microservices | React...

For **Spring Boot 3+**

```
<dependency>
   <groupId>org.springdoc</groupId>
   <artifactId>springdoc-openapi-starter-webmvc-ui</artifactId>
   <version>2.5.0</version>
</dependency>
```

Project Structure

src
 |
 +-- controller
 | StudentController
 |
 +-- service
 | StudentService
 |
 +-- entity
 | Student
 |
 +-- config
 | OpenApiConfig
 |
 +-- SpringBootApplication

Student Entity

```
@Entity
public class Student {

   @Id
   @GeneratedValue
   private Long id;

   private String name;

   private String email;

}
```

Ummed Singh
MTS@ByteXL | Ex - Nagarro | Java | Spring Boot | Microservices | React...

Student Controller

```
@RestController
@RequestMapping("/students")
public class StudentController {

   @PostMapping
   public Student createStudent(@RequestBody Student student) {
       return student;
   }

   @GetMapping
   public List<Student> getStudents() {

       return List.of(new Student(1L, "Ummed Singh", "ummed@gmail.com"));
   }
}
```

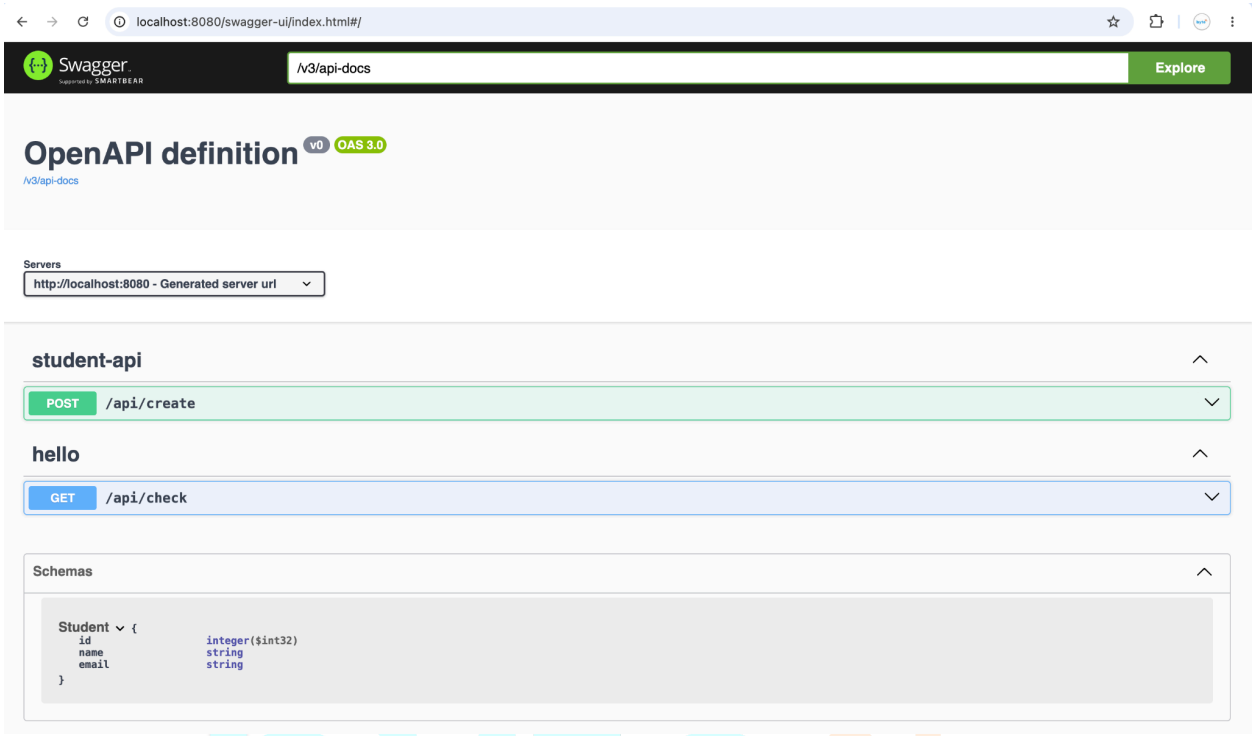
Run Application

After starting application: http://localhost:8080/swagger-ui.html

Or: http://localhost:8080/swagger-ui/index.html

Swagger UI appears automatically.

Ummed Singh
MTS@ByteXL | Ex - Nagarro | Java | Spring Boot | Microservices | React...



API Documentation Annotations

@Tag

Group APIs.

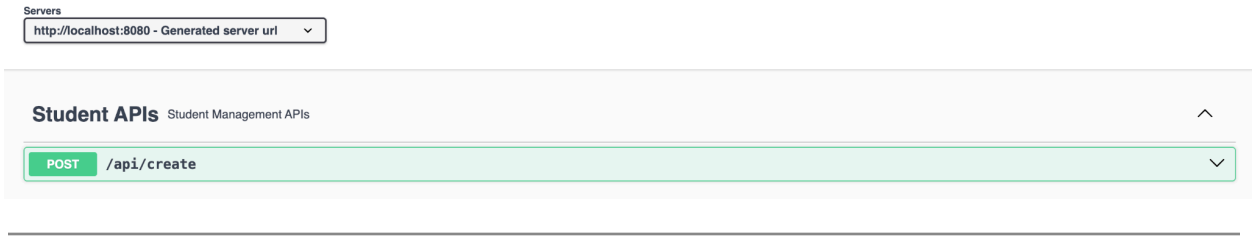
```java
import io.swagger.v3.oas.annotations.tags.Tag;

@Tag(name = "Student APIs",
    description = "Student Management APIs")
@RestController
@RequestMapping("/api")
public class StudentApi {

```

Swagger Output:

Ummed Singh
MTS@ByteXL | Ex - Nagarro | Java |
Spring Boot | Microservices | React...



@Operation

Describe endpoint.

```
@Operation(summary = "Create Student",
           description = "Creates a new student record")
@PostMapping
public Student createStudent(@RequestBody Student student){
   return student;
}
```

Swagger shows:

Create Student
Creates a new student record

@Parameter

Documents path/query parameters.

```
@GetMapping("/{id}")
public Student getStudent(@Parameter(description = "Student ID")
                          @PathVariable Long id){

   return service.findById(id);
}
```

@Schema

Documents model fields.

```
public class Student {

   @Schema(description = "Student Identifier", example = "1")
   private Long id;

   @Schema(description = "Student Name", example = "Ummed Singh")
   private String name;

   @Schema(description = "Student Email", example =
"ummed@gmail.com")
   private String email;
}
```

Swagger automatically displays examples.

@ApiResponse

Documents responses.

```
@Operation(summary = "Get Student")
@ApiResponses({@ApiResponse(responseCode = "200", description = "Student Found"),
               @ApiResponse(responseCode = "404", description = "Student Not Found")})
@GetMapping("/{id}")
public Student getStudent(@PathVariable Long id){

   return service.findById(id);
}
```

Endpoint Testing using Swagger UI

Suppose API:

```
@PostMapping
public Student createStudent(@RequestBody Student student){
   return student;
}
```

Swagger UI shows:

POST /students

Click: Try it out

Request:

```
{
 "name": "Ummed Singh",
 "email": "ummed@gmail.com"
}
```

Click: Execute

Swagger sends request and displays:

```
{
 "id": 1,
 "name": "Ummed Singh",
 "email": "ummed@gmail.com"
}
```

Swagger UI Architecture

Browser
|
|
Swagger UI
|
|
OpenAPI JSON
|
Spring Boot
Controllers
|
Service
|
Repository
|
Database

Flow:

Controller
|
Annotations
|
OpenAPI Generator
|
swagger.json
|
Swagger UI
|
Browser

Custom Swagger Configuration

Create configuration class.

```
@Configuration
public class OpenApiConfig {

   @Bean
   public OpenAPI customOpenAPI() {

       return new OpenAPI()
               .info(new Info().title("Student Management API")
                               .version("1.0")
                               .description("Spring Boot Swagger Example")
                               .contact(new Contact().name("Ummed Singh")
                                                     .email("ummed@gmail.com"))
               );
   }
}
```

Swagger UI Header:

Student Management API
Version 1.0
Spring Boot Swagger Example

Ummed Singh
MTS@ByteXL | Ex - Nagarro | Java |
Spring Boot | Microservices | React...

Configure Swagger Path

application.properties: springdoc.swagger-ui.path=/api-docs

Access: http://localhost:8080/api-docs

OpenAPI JSON Endpoint

Swagger automatically creates: http://localhost:8080/v3/api-docs

Output:

```
{
  "openapi":"3.0.1",
  "info":{
     "title":"Student API"
  }
}
```

This JSON is the actual OpenAPI specification.

Swagger with JWT Authentication

Without JWT:

Open Swagger
→ Execute API

With JWT:

Open Swagger
→ Authorize
→ Enter Token
→ Execute API

Configuration:

Ummed Singh
MTS@ByteXL | Ex - Nagarro | Java | Spring Boot | Microservices | React...

```java
@Bean
public OpenAPI customOpenAPI() {

   return new OpenAPI()
           .components(new Components()
                   .addSecuritySchemes("BearerAuth",
                           new SecurityScheme()
                                   .type(SecurityScheme.Type.HTTP)
                                   .scheme("bearer")
                                   .bearerFormat("JWT")
                   ))
           .addSecurityItem(
                   new SecurityRequirement()
                           .addList("BearerAuth")
           );
}
```

Swagger shows:

Authorize 🔒

Enter: Bearer eyJhbGciOi...

Now secured APIs can be tested.

Project Example

Imagine your Job Portal project.

APIs:

POST /auth/register
POST /auth/login

GET /jobs
POST /jobs

GET /profile
PUT /profile

Ummed Singh
MTS@ByteXL | Ex - Nagarro | Java |
Spring Boot | Microservices | React...

Swagger Benefits:

Frontend Team:
Can test APIs

QA Team:
Can validate endpoints

Developers:
Can understand APIs

Clients:
Can see available APIs

Interview Questions

What is Swagger?

Swagger is a tool used to generate interactive API documentation based on the OpenAPI specification.

What is OpenAPI?

OpenAPI is a standard specification used to describe REST APIs.

Which dependency is used in Spring Boot 3?

springdoc-openapi-starter-webmvc-ui

Swagger UI URL?

/swagger-ui/index.html

OpenAPI JSON URL?

/ v3 / api-docs

Why use Swagger?

Ummed Singh
MTS@ByteXL | Ex - Nagarro | Java | Spring Boot | Microservices | React...

✔ API Documentation
✔ API Testing
✔ Request/Response Examples
✔ Better Collaboration
✔ Reduced Development Time